CHANDIGARH ADMINISTRATION • MUNICIPAL CORPORATION CHANDIGARH

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Digital Public Infrastructure (DPI) • Open Hand Standard

CivicPulse Tricity NagrikSetu AI

Multilingual Citizen Infrastructure Intelligence Platform aggregating voice, text, and messaging into actionable municipal capital DPRs across Chandigarh, Mohali, and Panchkula.

Built by: **Sudhir Sharma (@satanic47)**

Google Gemini 1.5 Open-Source MIT 100% DPI Standard

Why Municipal Grievance Systems Are Broken

Linguistic Disconnect

70%+ of grassroots citizens in Chandigarh, Mohali, and rural fringe wards express complaints verbally in **Punjabi** or **Hindi**, but legacy portals only accept rigid English form dropdowns.

Fragmented Data Silos

Grievances are scattered across WhatsApp, phone helplines, municipal portals, and social media. No consolidated source of truth exists across Tricity municipal boundaries.

Budget Misalignment

Policymakers allocate multi-crore ward budgets based on historical inertia rather than real-time citizen demand hotspots and infrastructure hazard urgency.

CivicPulse Tricity bridges the gap between grassroots citizen voice and municipal capital works.

A Digital Public Good for Grassroots Governance

1. Zero-Friction Omnichannel Intake

Citizens simply send a voice note on WhatsApp or speak into the portal in **Punjabi, Hindi, or English**. Google Gemini transcribes, standardizes, geocodes, and triages the issue in sub-seconds.

2. Sector & Landmark Geo-Intelligence

Automated entity recognition maps colloquial landmarks ("Kiran Cinema", "Patel Market", "PU Gate 2") directly to Chandigarh Sector centroids, municipal wards, and administrative zones.

3. Demand Hotspot Clustering

Clustering algorithms aggregate individual complaints into high-density geographic demand clusters, computing localized urgency indices (1–10) and infrastructure health scores.

4. Municipal DPR Generator

Gemini AI analyzes hotspot density against ward census population and municipal budgets to generate executive Detailed Project Reports (DPR) for smart city engineering works.

Engineered under India's Digital Public Infrastructure (DPI) principles: modular, open, and scalable.

How Google AI Drives End-to-End Execution

Gemini 1.5 Multilingual Reasoning & Normalization

Ingests diverse dialect transcripts (Punjabi Gurmukhi, Devanagari Hindi, Hinglish), detects sentiment, and translates into canonical structured English schemas with zero semantic loss.

Urgency Scoring & Departmental Routing

Evaluates life-safety hazard parameters (e.g. exposed live electric wires = 10/10, burst water mains = 9/10), routing directly to the responsible wing: *Public Health, B&R Roads, Electricity, or Sanitation*.

Open-Weight Gemma Local Fallback for Sovereign Security

Designed to run locally on municipal servers via open-weight **Gemma 2**, ensuring government data sovereignty with 100% offline uptime and zero per-token cloud costs.

Proven 100% test pass rate across 21 automated pytest integration benchmarks.

Real-World Tricity Triage Results

Sector 22 Shastri Market — Urgency: 9/10

"ਸਾਡੇ ਸੈਕਟਰ 22 ਸ਼ਾਸਤਰੀ ਮਾਰਕੀਟ ਕੋਲ ਪਾਣੀ ਦੀ ਮੁੱਖ ਪਾਈਪ ਫਟ ਗਈ ਹੈ..."

AI Action: Triaged to Public Health & Water Supply. 18 citizen upvotes. Identified as Primary Hotspot #1.

Sector 35 Hotel Corridor — Urgency: 8/10

"सेक्टर 35 सी होटल स्ट्रिप के पीछे मुख्य सड़क पर बहुत बड़े गड्ढे हैं..."

AI Action: Triaged to B&R Roads. DPR generated for urgent bituminous resurfacing (₹14.4 Lakhs).

Phase 7 Mohali Market — Urgency: 8/10

"ਫੇਜ਼ 7 ਮੋਹਾਲੀ ਮਾਰਕੀਟ ਦੀਆਂ ਸਟਰੀਟ ਲਾਈਟਾਂ ਪਿਛਲੇ 10 ਦਿਨਾਂ ਤੋਂ ਬੰਦ ਹਨ..."

AI Action: Cross-boundary routing to SAS Nagar Municipal Corporation Electricity Wing.

Sector 15 Patel Market — Urgency: 10/10

"सेक्टर 15 पटेल मार्केट के पास बिजली के खंभे से नंगी तारें लटक रही हैं..."

AI Action: Maximum priority alert triggered for immediate lineman dispatch within 2 hours.

Live interactive portal available at localhost:8000 with real-time Leaflet GIS mapping.

From Chandigarh to 4,000+ Urban Local Bodies

Deploy in 5 Minutes

Packaged in standard Docker containers with zero external proprietary database lock-in. Any municipal IT team can spin up a dedicated instance with a single command.

22 Scheduled Languages

Google Gemini's multi-lingual foundation enables instant scaling from Punjabi & Hindi in Chandigarh to Tamil, Bengali, Telugu, and Marathi across India.

Smart City Mission Fit

Seamlessly integrates into Integrated Command and Control Centres (ICCC) through open REST APIs and standard GeoJSON interfaces.

Empowering Communities Through Open AI

CivicPulse Tricity transforms passive citizens into active co-creators of their city's infrastructure. By merging multilingual voice AI with democratic budget transparency, we ensure no citizen request is ever lost in the bureaucratic void.

View GitHub Repository

Open-Source MIT Licensed